

Ch 5 Integrals

A great achievement of classical geometry was obtaining formulas for the areas and volumes of triangles, spheres, and cones. In this chapter we develop a method, called *integration*, to calculate the areas and volumes of more general shapes. The *definite integral* is the key tool in calculus for defining and calculating areas and volumes.

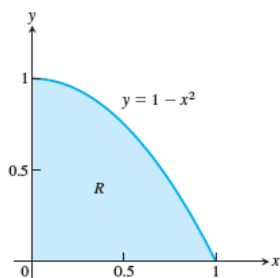
Like the derivative, the definite integral is defined as a limit. The definite integral is a limit of increasingly fine approximations. We also show that the process of computing these definite integrals is closely connected to finding antiderivatives. This is one of the most important relationships in calculus; it gives us an efficient way to compute definite integrals, providing a simple and powerful method that eliminates the difficulty of directly computing limits of approximations. This connection is captured in the Fundamental Theorem of Calculus.

5.1 Area and Estimating with Finite Sums

① Area



$$\text{Area} = hl$$



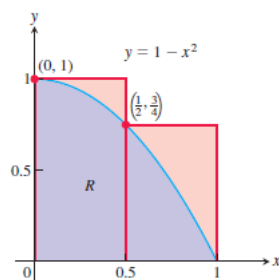
Suppose we want to find the area of the shaded region R that lies above the x -axis, below the graph of $y = 1 - x^2$, and between the vertical lines $x = 0$ and $x = 1$.

Unfortunately, there is no simple geometric formula for calculating the areas of general shapes having curved

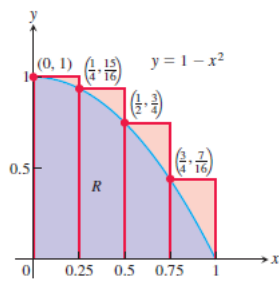
boundaries like the region R . How, then, can we find the area of R ?

While we do not yet have a method for determining the exact area of R , we can approximate it in a simple way.

(a) upper sum

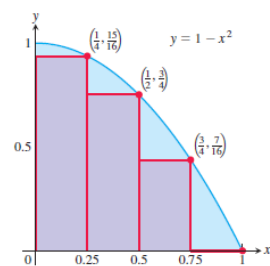


$$\text{Area} \approx \frac{1}{2} \cdot 1 + \frac{1}{2} \cdot \frac{3}{4} = \frac{7}{8} = 0.875$$



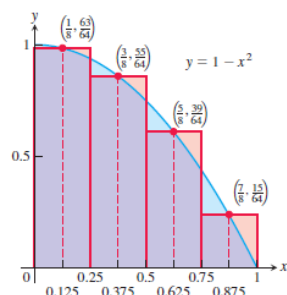
$$\text{Area} \approx \frac{1}{4} \cdot 1 + \frac{1}{4} \cdot \frac{15}{16} + \frac{1}{4} \cdot \frac{3}{4} + \frac{1}{4} \cdot \frac{7}{16} = \frac{25}{32} = 0.78125$$

(b) lower sum



$$\text{Area} \approx \frac{1}{4} \cdot \frac{15}{16} + \frac{1}{4} \cdot \frac{3}{4} + \frac{1}{4} \cdot \frac{7}{16} + \frac{1}{4} \cdot 0 = \frac{17}{32} = 0.53125$$

(c) midpoint sum



$$\text{Area} \approx \frac{1}{4} \cdot \frac{63}{64} + \frac{1}{4} \cdot \frac{55}{64} + \frac{1}{4} \cdot \frac{39}{64} + \frac{1}{4} \cdot \frac{15}{64} = 0.671875$$

TABLE 5.1 Finite approximations for the area of R

Number of subintervals	Lower sum	Midpoint sum	Upper sum
2	0.375	0.6875	0.875
4	0.53125	0.671875	0.78125
16	0.634765625	0.6669921875	0.697265625
50	0.6566	0.6667	0.6766
100	0.66165	0.666675	0.67165
1000	0.6661665	0.66666675	0.6671665

$$\text{Area} = \frac{2}{3} \approx n = 1000 \text{ midpoint sum.}$$

Summary

The area under the graph of a positive function, can be approximated by finite sums constructed in a certain way. First, we subdivide the interval into subintervals, treating f as if it were constant over each subinterval. Then we multiply the width of each subinterval by the value of f at some point within it, and add these products together. If the interval $[a, b]$ is subdivided into n subintervals of equal widths $\Delta x = \frac{b-a}{n}$, and if $f(c_k)$ is the value of f at the chosen point c_k in the k -th subinterval, this process gives a finite sum of the form

$$f(c_1) \Delta x + f(c_2) \Delta x + f(c_3) \Delta x + \cdots + f(c_n) \Delta x.$$

The choices for the c_k could maximize or minimize the value of f in the k -th subinterval, or give some value in between. The true value lies somewhere between the approximations given by upper sums and lower sums. In the examples that we looked at, the finite sum approximations improved as we took more subintervals of thinner width.

Example 1 Use finite approximations to estimate the area under the graph of the function $f(x) = x^2$ between $x = 0$ and $x = 1$ using

- a lower sum with two rectangles of equal width.
- an upper sum with two rectangles of equal width.